B.TECH CYBER FORENSICS CAPSTONE

Android Deleted Data Recovery & Evidence Reconstruction Framework

A deterministic, explainable pipeline for recovering, carving and correlating deleted evidence from Android SQLite databases — every result traced to a byte offset, never a guess.

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THE PROBLEM

On Android, deleted data is often still on disk

Messages, call logs, contacts and browsing history live in SQLite databases. A **DELETE** does not erase bytes — it unlinks a cell and frees the space, which stays readable until something else reuses it. Existing acquisition tools show what is *active*; almost none reconstruct what was removed — and none of them explain *why* a recovered value should be trusted.

Criminal investigation

Reconstructing a suspect’s deleted communications with full provenance

Incident response

Timelines of activity across active and deleted app data

Regulatory & civil discovery

Results a court can trust: reproducible, explained, never guessed

DESIGN PRINCIPLES

Six rules that are never broken

Accuracy over features

A correct partial answer beats an impressive wrong one

Provenance on everything

Every result traces to a byte offset, a hash and a method

Explainability

No black-box scoring — every confidence level ships its reasoning

Reproducibility

Same evidence, same bytes, same result — every single run

Evidence is immutable

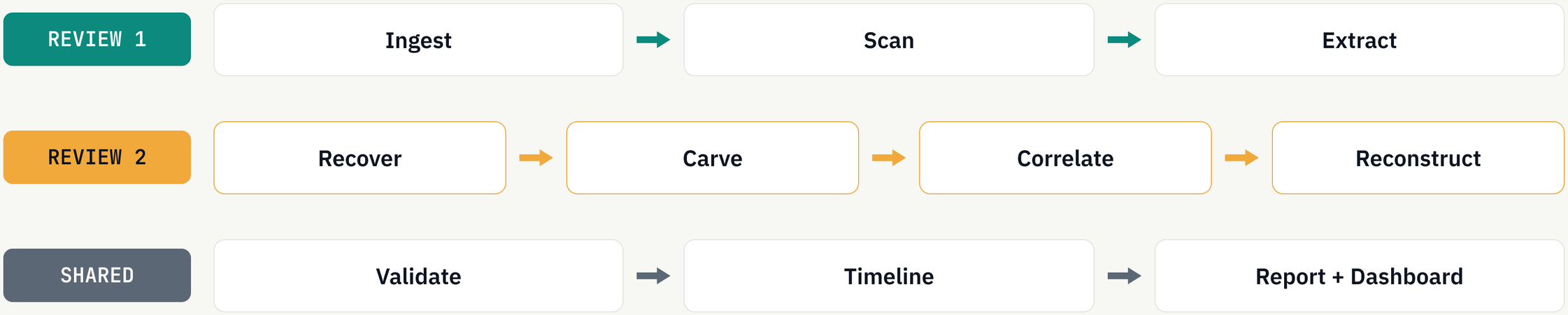
Originals read once, read-only; recovery works on scratch copies

Never fabricate

UNKNOWN means unknown — never guessed, never defaulted

SYSTEM DESIGN

Two reviews, one pipeline



Recovery is opt-in but on by default: **analyze** runs the full pipeline; **--no-recover** restores exactly Review 1’s behaviour, unchanged either way.

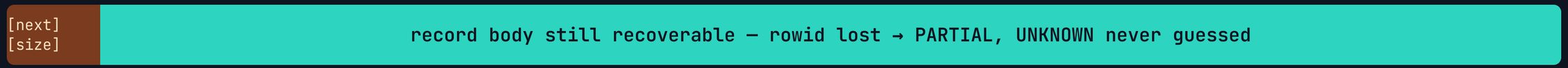
HOW RECOVERY WORKS

What survives a DELETE

Live cell — every byte intact



After DELETE — SQLite overwrites only the first 4 bytes (freeblock header)



Freeblocks & freelist

The main file’s own released cells and pages

WAL frames

Committed, uncommitted and stale-generation page history

Rollback journal

Pre-transaction page images from an interrupted write

THE RECOVERY ENGINE

Byte-for-byte, page by page

Every page of the main file, every WAL frame and every rollback-journal page is scanned for byte sequences that parse as a record of a known table — live cell extents are excluded first, so nothing already visible is ever re-reported.

RECOVERED

The whole cell survived — rowid, every field intact and structurally valid

PARTIAL

A freeblock header overwrote the first bytes — rowid unknown; lost columns marked UNKNOWN, never guessed

CORRUPTED

Structure recoverable, but a field fails validation — kept, and marked as untrustworthy

Measured on the 28-condition recovery matrix
511 physically recoverable rows, ground truth from re-encoded bytes

99.22%

RAW CARVING

Patterns in bytes no parser reads

- Phone numbers (E.164, NANP)
- URLs & Android URIs
- E-mail addresses
- Timestamps (Unix, WebKit, ISO)
- SMS-like text
- Loose SQLite record shapes

Status is always the literal **CARVED CANDIDATE**. Confidence can never be HIGH. Never presented as evidence until an independent source corroborates it.

CORRELATION & CONFIDENCE

Linking evidence, scoring it honestly

Correlation graph

Person, PhoneNumber, Message, Call, Contact, Database, URL and more — linked by SENT, RECEIVED, CALLED, CONTACT_OF, CORRELATES_WITH.

```
phone --CONTACT_OF--> contact
```

Corroboration only from an **independent** source — a different database, never the same file's data seen twice.

Confidence scorer

Schema match · structural validity · provenance · corroboration · completeness · timestamp plausibility.

LOW | MEDIUM | HIGH

Deterministic. No machine learning. The same bytes score the same way, every time.

USING THE TOOL

One CLI, one dashboard, four report formats

terminal

```
$ android-forensics analyze case001
Artifacts (ACTIVE): 346
Deleted-data recovery (Review 2)
  reconstructed → RECOVERED: 8,218  PARTIAL: 26
Carved candidates: 177  ·  Correlation edges: 9,079
$ android-forensics dashboard case001
Dashboard: http://127.0.0.1:8765/
```

Read-only dashboard

FastAPI REST API + static frontend. Never runs a stage — only shows what already ran.

HTML | JSON | CSV | PDF

Every report section shows real recovered / carved / correlated content when that stage ran — or an explicit “NOT PERFORMED” box when it did not. Never both.

EVALUATION — RECOVERY

Measured against physical ground truth

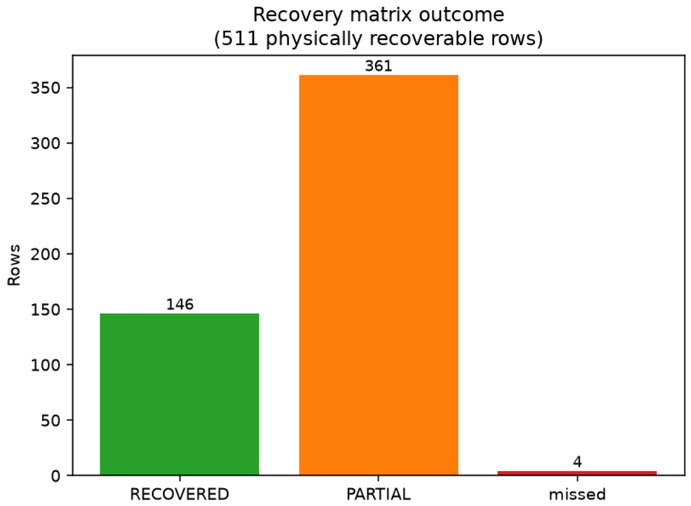
99.22%

recall — 507 of 511 recoverable rows

0
false positives

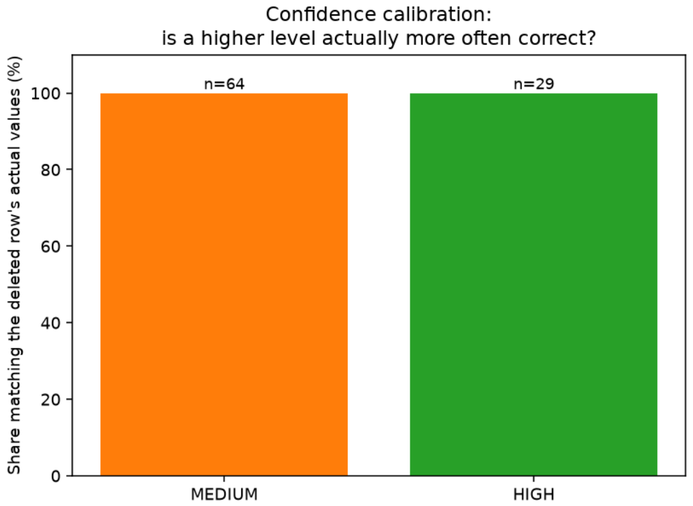
28
conditions tested

Ground truth: every deleted row is re-encoded exactly as SQLite would write it, then searched for in the file's real bytes — never predicted by the algorithm being measured.



EVALUATION — CONFIDENCE

Does HIGH actually mean more likely correct?



29
HIGH → 100% correct

64
MEDIUM → 100% correct

Every recovered artifact scored HIGH or MEDIUM was checked against the actual deleted values it claims to reconstruct — not just its own internal structure.

Measured on a 3-seed synthetic sample — see limitations for what this does and does not prove.

QUALITY

1,482 tests, 95.7% coverage

1,482
tests — 1,481 pass, 1 platform skip

95.7%
line coverage of the source tree

0
evidence-hash mismatches after a full run

0
INVALID validation results

Phase 17 acceptance run — the literal sequence executed for this project:

```
init-case → ingest → scan → analyze → validate → report → dashboard
```

Every one of the 16 original evidence files was re-hashed after the run and verified byte-identical to the ingest manifest.

LIMITATIONS

What this build does not claim

- Recovery does not attempt tables with fewer than 3 columns, **WITHOUT ROWID** tables, or rowids needing more than 6 varint bytes
- A journal-sourced record’s checksum is not re-derived at reconstruct time — it defaults to verified, unlike WAL frames
- The synthetic lab’s genuine deletions are limited to two of the four evidence databases
- Carving matches a byte *shape*, not meaning — a bare digit run is never called a phone number
- Confidence calibration was measured on a small synthetic sample — 93 artifacts across three seeds

PHYSICALLY IMPOSSIBLE

No tool can recover this

Full-disk encryption, no key

Modern Android encrypts user data with keys bound to the lock-screen credential. No key extraction is attempted, by design.

TRIM & wear-levelling

Flash storage erases freed blocks in the background, often before an image is even taken.

Genuinely overwritten pages

Reused freelist pages, rewritten WAL frames, a vacuumed database — the bytes are simply gone.

Data never acquired

A logical acquisition omits what a physical/full-filesystem one contains. What was never acquired cannot be analysed.

The absence of a recovered or carved result is **never** evidence that the data never existed — this sentence is printed on every report this tool generates.

NEXT STEPS

Where this goes next

01 Re-verify journal-record checksums at reconstruct time, closing the one provenance gap WAL already fills

02 Genuine deletions in the WAL-mode and multi-table databases of the synthetic lab

03 A larger confidence-calibration corpus — more seeds, and eventually real device acquisitions

04 E.164 phone-number parsing with a real numbering-plan database

05 Validation against real, authorised device acquisitions across OEMs and Android versions

THANK YOU

Questions?

Android Deleted Data Recovery & Evidence Reconstruction Framework — 1,482 tests, 95.7% coverage, 99.22% measured recovery recall, zero fabricated numbers.

[Your Name] · [Email] · Repository: [link]